

SANDSTONE 6 W, MN, USA

WMO: 726460

Lat: 46.1135N

Lon: 92.9936W

Elev: 1130

StdP: 14.11

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 54932

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.9	-12.9	-24.3	1.5	-17.1	-20.0	1.9	-11.9	10.5	12.9	9.5	12.7	1.3	N/A	0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	85.2	71.0	82.4	69.0	80.0	67.1	74.2	82.2	71.8	79.6	69.8	77.2	5.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.4	121.0	79.9	69.1	111.5	76.7	67.2	104.1	74.4	38.5	81.9	36.3	79.6	34.5	77.5	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.0	8.7	7.8	-25.8	90.0	5.4	2.4	-29.6	91.7	-32.8	93.1	-35.8	94.5	-39.7	96.2
			-24.2	78.4	4.1	2.3	-27.2	80.1	-29.6	81.4	-31.9	82.7	-34.8	84.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.5	10.7	14.5	28.6	40.9	53.6	62.6	68.0	65.6	58.6	45.5	31.1	16.8
	DBStd	22.15	13.99	12.24	13.05	9.10	8.23	6.10	5.15	5.19	7.90	8.39	10.80	12.06
	HDD50	5045	1218	994	672	296	53	2	0	0	16	193	571	1030
	HDD65	8855	1683	1414	1130	723	367	113	29	57	220	606	1018	1495
	CDD50	1946	0	0	7	23	164	380	558	483	275	52	4	0
	CDD65	280	0	0	0	0	13	41	121	75	29	1	0	0
	CDH74	2670	0	0	1	5	232	475	1050	632	252	23	0	0
	CDH80	536	0	0	0	0	63	92	217	119	44	1	0	0
(14) Wind	WSAvg	3.3	3.3	3.6	3.5	4.2	4.0	3.5	2.8	2.4	2.6	3.1	3.3	3.3
(15) Precipitation	PrecAvg	31.6	0.9	0.9	1.5	2.8	3.2	4.6	4.3	4.1	3.3	3.2	1.8	1.1
	PrecMax	39.6	2.5	2.5	2.6	8.1	6.5	7.5	7.0	9.3	6.1	6.7	5.2	2.3
	PrecMin	20.8	0.0	0.1	0.7	0.6	1.1	2.4	0.9	1.0	0.8	0.8	0.1	0.3
	PrecStd	5.0	0.7	0.7	0.6	1.7	1.3	1.3	1.7	2.0	1.7	1.6	1.5	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.2	48.7	69.0	73.5	86.8	87.0	88.6	87.0	84.9	77.7	64.0	44.4
		MCWB	36.3	42.9	58.5	56.0	64.1	71.0	76.2	73.7	70.0	60.4	52.7	40.9
	2%	DB	36.8	39.8	59.0	68.0	80.6	82.4	84.9	83.4	80.4	70.7	57.1	39.0
		MCWB	33.4	34.9	50.0	51.2	61.2	66.9	72.9	71.5	68.5	56.3	49.1	36.1
	5%	DB	33.6	35.6	51.6	62.7	75.1	79.3	82.3	80.3	76.7	64.3	51.3	35.2
		MCWB	31.6	31.7	44.2	48.3	58.7	64.6	70.1	68.3	65.1	53.1	44.4	33.5
	10%	DB	30.4	31.9	45.6	58.0	70.6	76.5	79.9	77.6	72.8	59.2	46.6	33.0
		MCWB	28.3	29.1	40.1	45.6	56.1	63.3	68.3	66.5	63.4	51.4	41.6	31.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.0	43.2	61.1	57.4	68.3	74.8	79.5	75.5	74.3	62.5	54.8	43.8
		MCDB	41.4	51.0	71.4	72.2	81.6	83.8	87.5	84.2	81.3	75.7	60.6	45.0
	2%	WB	34.3	37.0	51.7	52.6	64.5	70.8	75.1	72.7	70.7	58.8	49.0	37.0
		MCDB	36.7	40.8	57.4	64.2	73.8	79.6	82.8	80.8	77.3	67.2	56.1	39.2
	5%	WB	32.0	32.5	45.0	49.1	62.1	67.6	72.6	70.6	68.1	55.6	45.1	34.4
		MCDB	34.5	35.8	51.6	60.8	71.1	75.7	80.5	77.8	74.1	63.9	51.0	36.0
	10%	WB	28.8	29.7	40.1	46.1	59.3	65.4	70.3	68.5	65.4	52.6	41.7	32.4
		MCDB	30.8	32.4	45.9	56.2	68.5	73.0	77.9	75.0	72.1	58.9	46.4	34.0
(35) Mean Daily Temperature Range	5% DB	MDBR	17.0	18.5	19.0	20.3	23.6	21.4	20.5	20.8	20.9	17.9	15.0	13.8
		MCDBR	17.2	19.3	24.4	31.0	31.4	25.7	22.3	23.4	25.1	26.9	22.8	13.8
		MCWBR	14.2	13.8	15.5	16.9	15.1	13.4	11.6	12.5	13.3	15.3	15.2	11.6
	5% WB	MCDBR	16.7	18.3	23.0	27.1	25.2	22.5	20.8	20.9	21.1	24.1	22.2	12.2
		MCWBR	14.0	13.5	15.8	16.2	14.3	13.3	12.2	12.4	12.7	14.6	15.2	10.4
(40) Clear-Sky Solar Irradiance	taub		0.250	0.261	0.307	0.362	0.391	0.379	0.378	0.389	0.362	0.311	0.284	0.258
	taud		2.399	2.376	2.371	2.330	2.293	2.356	2.384	2.369	2.418	2.555	2.517	2.427
	Ebn at Noon		277	295	293	284	278	281	280	272	271	273	262	260
	Edh at Noon		22	28	33	37	40	38	36	36	31	23	20	19
(44) All-Sky Solar Radiation	RadAvg		516	868	1219	1474	1692	1854	1969	1643	1254	775	485	363
	RadStd		56	77	112	141	132	130	73	124	90	103	47	39

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page